Problem Set 2

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Before you begin: Happy Gandhi Jayanti! The problems are roughly arranged in order of difficulty. Nevertheless, I have labelled them with quotes from some of the members of our class. To maximize fun, use algebra, limit laws, elementary trigonometric identities, and the squeeze theorem; and try to avoid L'Hôpital's rule, derivatives, or Taylor series.

1 "Trivial"

Q1 Making sure to cancel meticulously, evaluate

$$\text{(a)} \quad \lim_{x\to 2} \frac{x^5 - 32}{x - 2}, \qquad \text{(b)} \quad \lim_{x\to 1} \frac{x^4 - 1}{x^3 - 1}.$$

(c) More generally, find

$$\lim_{x\to a} \frac{x^n - a^n}{x - a}$$

where $n \in \mathbb{N}, a \in \mathbb{R}$ and explain "crisply" why cancellation is possible.

Q2 Using $\lim_{t\to 0} \frac{\sin t}{t} = 1$, evaluate

$$\text{(a)} \quad \lim_{x\to 0} \frac{\sin(3x)}{\tan(2x)}, \qquad \text{(b)} \quad \lim_{x\to 0} \frac{1 - \cos(3x)}{x^2},$$

$$\text{(c)} \quad \lim_{x\to 0} \frac{\sin(2x)\sin(5x)}{x\sin(3x)}.$$

Q3 Prove directly from the ε–δ definition that

$$\lim_{x\to 2}(x^2 + x) = 6.$$

Give an explicit choice of δ in terms of ε. Then morph your argument to prove

$$\lim_{x\to a}(x^2 + x) = a^2 + a$$

for every $a \in \mathbb{R}$.

2 "Well-*Rounded* (Get it?)"

Q4 (a) Prove the squeeze theorem using the ε–δ definition.

(b) Determine whether each limit exists with reasoning.

$$\lim_{x\to 0} \sin(1/x), \qquad \lim_{x\to 0} x\sin(1/x), \qquad \lim_{x\to 0} x\sin(1/x^2).$$

(c) For which real numbers α does

$$\lim_{x\to 0} |x|^{\alpha} \sin(1/x)$$

exist? Find its value whenever it does.

Q5 (a) Prove directly from the definition that

$$\lim_{x\to+\infty} \frac{3x+1}{x^2+1} = 0.$$

Q6 Suppose $f(x) \to L$ and $g(x) \to M$ as $x \to a$. A "child" writes

$$|f(x)g(x) - LM| \leq |g(x)||f(x) - L| + |L||g(x) - M|$$

and chooses both errors to be smaller than

$$\frac{\varepsilon}{2(|M|+1)}.$$

After also ensuring $|g(x)| < |M| + 1$, the "child" concludes that the result is less than ε.

(a) Explain why this estimate does not justify the conclusion for arbitrary L, M.

(b) Rectify the proof, without assuming any ordering between $|L|$ and $|M|$.

Q7 Suppose $f(x) \to L \neq 0$ as $x \to a$.

(a) Show that, sufficiently close to a,

$$|f(x)| > \frac{|L|}{2}.$$

(b) Use this to prove directly that

$$\lim_{x\to a} \frac{1}{f(x)} = \frac{1}{L}.$$

(c) Let's say $f(x) \to 0$, with $f(x) \neq 0$ in some neighbourhood of a. Can $1/f(x)$ have a finite limit? Prove or disprove it.

3 "The only kinds of lemons we're ever gonna get"

Q8 Let p be a nonzero real polynomial and let a be a root of p.

(a) Prove that

$$\lim_{x\to a} \frac{1}{p(x)}$$

does not exist.

(b) Write

$$p(x) = (x-a)^k q(x), \qquad q(a) \neq 0,$$

where $k \geq 1$ is the multiplicity of the root. Determine the behaviour of $1/p(x)$ on each side of a, in terms of the sign of $q(a)$ and the parity of k.

(c) Give an example where both sides tend to $+\infty$, and another where one goes to $+\infty$ and the other $-\infty$.

Q9 For $(x, y) \neq (0, 0)$, define

$$F(x, y) = \frac{x^2y + xy^2}{x^2 + y^2}.$$

(a) Prove that $F(x, y) \to 0$ as $(x, y) \to (0, 0)$.

(b) Give an explicit $\delta(\varepsilon)$ such that

$$0 < \sqrt{x^2 + y^2} < \delta \quad \Longrightarrow \quad |F(x, y)| < \varepsilon.$$

(c) Does the same conclusion hold if the numerator is replaced by xy? Justify your answer.

Q10 For $(x, y) \neq (0, 0)$, define

$$G(x, y) = \frac{x^2y}{x^4 + y^2}.$$

(a) Show that $G(x, y) \to 0$ along every straight line through the origin, including the vertical line.

(b) Investigate the paths $y = cx^2$, where $c \in \mathbb{R}$. Can you find the full 2-variable limit?

(c) For each positive integer m, determine whether

$$\lim_{(x,y)\to(0,0)} \frac{x^m y}{x^4 + y^2}$$

exists.

4 "Batshit Crazy"

Q11 For $(x, y) \neq (0, 0)$, define

$$F(x, y) = \frac{x(y \sin x)}{x^4 + (y - \sin x)^2}$$

(a) Prove, disprove or salvage if possible (PODASIP): $F(x, y)$ tends to 0 for every straight line through the origin.

(b) Find every real a such that there is a continuous path $(x(t), y(t))$ approaching $(0, 0)$ as $t \to 0^+$, along which $F(x(t), y(t)) \to a$. Hence decide whether a 2-variable limit exists.

P.S. Sorry if these questions feel like the classic STEP template, that's what I've been doing nowadays so it's all I can write.